

Lab 9: Histogram Equalization

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A 2D kernel counts 256 grayscale bins using atomic additions. The histogram is reset before each timed run. The CPU builds the CDF lookup table using the first occupied CDF value; a CUDA map applies it. Constant images are preserved. The notebook validates counts and mapping, saves histogram CSV data, and plots before/after histograms. Histogram and mapping times are separate; CDF construction is on the CPU.

Kernel timing uses `time.time()` after JIT warm-up and CUDA synchronization, averaged over three GPU runs. Device allocation and host/device copies are excluded. CPU measurements are single runs. Speedup is relative to the stated CPU implementation. Actual measurements appear after running the notebook on Colab; no timings are assumed here.

histogram_ms	equalization_ms	pixels	max_error
0.120481	0.0927448	65536	0

